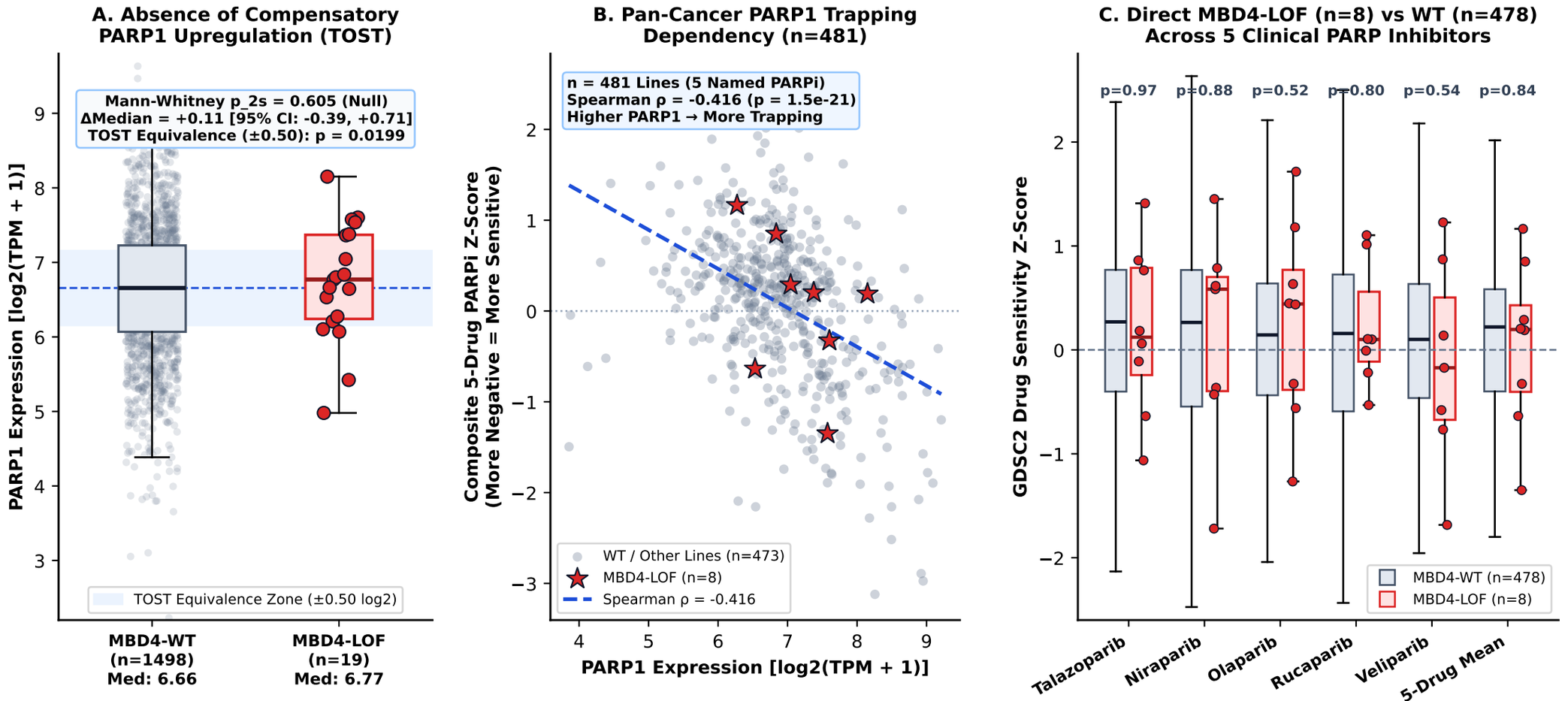

A. Absence of Compensatory PARP1 Upregulation (TOST)
Mann-Whitney p_2s = 0.605 (Null)
ΔMedian = +0.11 [95% CI: -0.39, +0.71]
TOST Equivalence (±0.50): p = 0.0199
PARP1 Expression [log2(TPM + 1)]
TOST Equivalence Zone (±0.50 log2)
MBD4-WT (n=1498) Med: 6.66
MBD4-LOF (n=19) Med: 6.77
B. Pan-Cancer PARP1 Trapping Dependency (n=481)
n = 481 Lines (5 Named PARPi)
Spearman ρ = -0.416 (p = 1.5e-21)
Higher PARP1 → More Trapping
Composite 5-Drug PARPi Z-Score (More Negative = More Sensitive)
PARP1 Expression [log2(TPM + 1)]
WT / Other Lines (n=473)
MBD4-LOF (n=8)
Spearman ρ = -0.416
C. Direct MBD4-LOF (n=8) vs WT (n=478) Across 5 Clinical PARP Inhibitors
p=0.97
p=0.88
p=0.52
p=0.80
p=0.54
p=0.84
GDSC2 Drug Sensitivity Z-Score
MBD4-WT (n=478)
MBD4-LOF (n=8)
Talazoparib
Niraparib
Olaparib
Rucaparib
Veliparib
5-Drug Mean